

AE-1: ECG Anomaly Detection with Autoencoders

Data Mining — Deep Learning, Foundational Project (20 pts)

Sobhan Moghimi

Amirhossein Azarpour

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Instructor: Dr. Hadi Farahani

Abstract

This report detects anomalous heartbeats on ECG5000 with reconstruction autoencoders trained on normal beats only. Class 1 is treated as normal; classes 2–5 are anomalies. The original contribution is a leakage-safe Conv1D vs. LSTM autoencoder comparison under the same split, scaler, loss, and two justified thresholds: a normal-validation 95th percentile (no anomaly labels) and an F1-optimal cut on a labeled calibration set. An untouched stratified test set of 1000 beats is scored once. The LSTM autoencoder is the better neural detector (test F1 0.970, AP 0.962). The smaller Conv1D model is fast but overlaps normal and abnormal errors; its unsupervised P95 threshold collapses to recall 0.24. A One-Class SVM stretch baseline slightly edges LSTM on F1 (0.972) while remaining weaker on ranking metrics. Accuracy is not reported as a headline number: the test set is 58% normal.

1 Problem and data

ECG5000 contains 5000 univariate beats of length 140, five morphology classes [1, 2, 3]. The task is one-class anomaly detection: train on normal beats, then flag a test beat if reconstruction error exceeds a threshold. Because the classes are imbalanced, we report precision, recall, and F1 for the anomaly class, plus average precision and ROC AUC. Accuracy alone would be misleading.

Source. Loaded with `aeon.datasets.load_classification('ECG5000')` (aeon 1.5.0), combining the archive train and test files. Shape (5000, 140); labels {1, 2, 3, 4, 5}. We use the conventional names Normal, R-on-T PVC, PVC, SP/EB, and Unclassified.

Split (leakage-safe). A stratified 20% final test is carved out first and not used for scaling, training, early stopping, or threshold choice. From the remainder, 20% is a labeled calibration set. Remaining class-1 beats are split 80/20 into normal train and normal validation. A feature-wise `StandardScaler` is fit on the 1494 normal training beats only.

Split	<i>n</i>	Normal	Anom.	C2	C3	C4	C5
Normal train	1494	1494	0	0	0	0	0
Normal val	374	374	0	0	0	0	0
Labeled calibration	800	467	333	283	15	31	4
Final test	1000	584	416	353	19	39	5

Anomalies on test are dominated by R-on-T PVC (353/416). PVC and unclassified supports are small, so per-type recall is noisy there.

2 Method

Both autoencoders take a beat (140, 1) and minimize MSE reconstruction. Training uses only normal train beats. Early stopping and `ReduceLROnPlateau` watch *normal-validation* loss (patience 8). Adam 10^{-3} , batch 64, at most 60 epochs, seed 42, TensorFlow 2.20 on one GPU. Reported run: `FAST_MODE=False`.

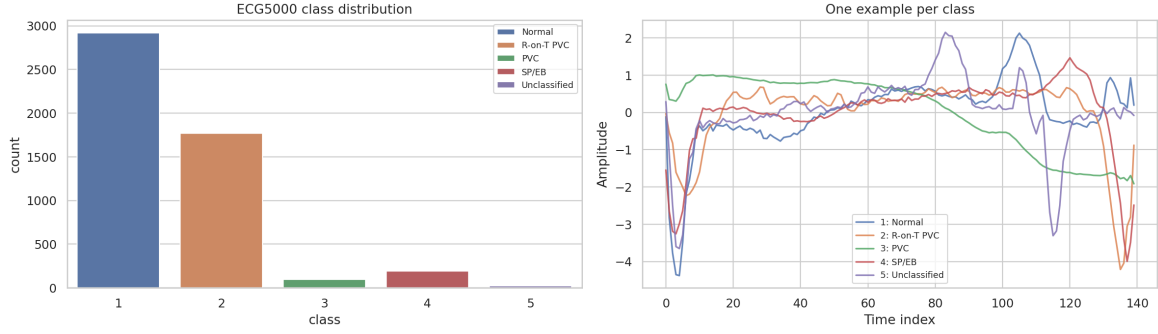


Figure 1: Class counts and representative waveforms. Class 1 (normal) is the majority; class 2 dominates the anomalies.

Conv1D AE (7841 parameters, 24 s). Two conv–pool stages (32 then 16 filters, kernels 7 and 5) compress to a 35×16 latent map; symmetric conv–upsample reconstructs. Local filters are a natural inductive bias for QRS-like motifs.

LSTM AE (28,273 parameters, 53 s). An LSTM encodes the beat into a 48-dimensional vector, `RepeatVector` expands it, a second LSTM plus a time-distributed dense layer reconstructs. The bottleneck is a single vector, not a downsampled sequence.

The score is mean squared error over the 140 samples. Larger score $\Rightarrow$ more anomalous.

Two thresholds, frozen before test.

- **Normal-val P95.** 95th percentile of scores on held-out normal beats. No anomaly labels. Nominally $\approx 5\%$ of similar normals should alarm.
- **Calibration F1.** Threshold maximizing F1 on the labeled calibration split. Uses representative anomalies, so it can overfit prevalence and type mix.

Calibration F1 on the labeled subset was 0.816 (Conv1D) and 0.951 (LSTM). Those cuts, not test F1, are what we freeze.

3 Original contribution

The project card asks for a scoring or architecture choice beyond a stock autoencoder [1]. Ours is a **controlled Conv1D AE vs. LSTM AE** comparison: same data, scaler, MSE, optimizer family, early-stopping rule, score, and threshold policies. Parameter counts are disclosed rather than forced equal; the question is which complete design is preferable, not a capacity-matched ablation.

A second, explicit piece is the **threshold study**: unsupervised P95 vs. labeled F1, reported on the same untouched test set. Hypothesis before looking at test numbers: LSTM should separate better if long-range beat context matters; Conv1D should be close if local motifs suffice, and then we would prefer it for size and speed. Labeled calibration should beat P95 on F1 when the calibration mix matches test; P95 should be safer when labels are unavailable, at some recall cost.

4 Results

Figure 2 shows training. Conv1D validation MSE falls to about 0.015; LSTM stays near 0.47. Those losses are not comparable across architectures (different residual scales). What matters is whether *normal and abnormal scores separate*.

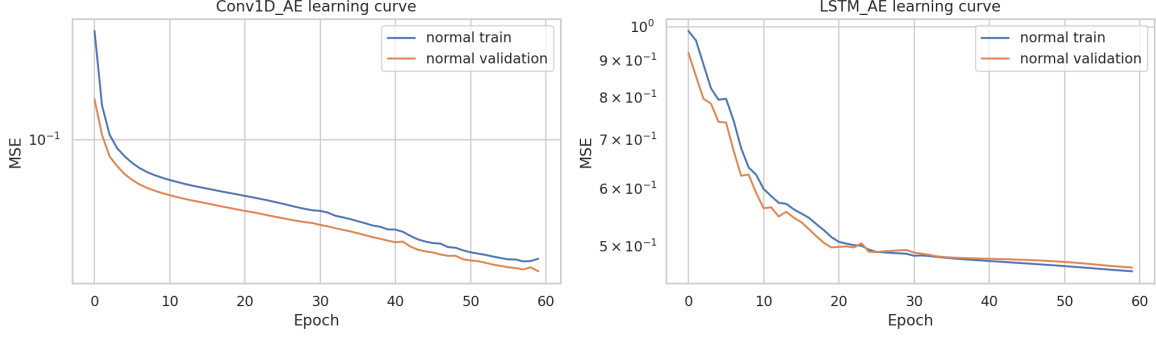


Figure 2: Normal-train and normal-validation MSE. Checkpoints restore the best validation epoch.

Main table. Table 1 is the frozen-test result. Under the same calibration-F1 rule, LSTM F1 is 0.970 vs. Conv1D 0.819. The gap is precision, not recall: Conv1D catches most anomalies ($R = 0.959$) but raises 159 false alarms on 584 normals. LSTM has 14 FP and 11 FN. Ranking metrics, which do not depend on the cut, agree: LSTM AP 0.962 / AUC 0.988 vs. Conv1D 0.714 / 0.877.

The unsupervised P95 rule makes the architecture gap even clearer. LSTM P95 still has F1 0.965 and recall 0.995 (2 misses). Conv1D P95 is specific (0.933) but almost blind (recall 0.238, F1 0.357): most anomalies sit below the normal tail. So Conv1D reconstructs many abnormal beats well enough to look normal — exactly the reconstruction paradox in numbers.

Table 1: Untouched test set ($n = 1000$, 416 anomalies). Bold is best neural result in the column among calibration-F1 rows. Stretch baselines are shown for context.

Model	Threshold	P	R	F1	AP	AUC	FP / FN
LSTM AE	calib. F1	0.967	0.974	0.970	0.962	0.988	14 / 11
Conv1D AE	calib. F1	0.715	0.959	0.819	0.714	0.877	159 / 17
LSTM AE	val P95	0.937	0.995	0.965	0.962	0.988	28 / 2
Conv1D AE	val P95	0.717	0.238	0.357	0.714	0.877	39 / 317
One-Class SVM	calib. F1	0.952	0.993	0.972	0.948	0.986	21 / 3
Isolation Forest	calib. F1	0.901	0.983	0.940	0.866	0.962	45 / 7

Which anomalies are missed. Table 2 and Figure 6 break recall down by type at the calibration-F1 cut. LSTM detects 346/353 R-on-T PVCs, 17/19 PVCs, 37/39 SP/EB, and 5/5 unclassified. Conv1D is similar on recall once the threshold is lowered, but that is the setting that floods false positives. Under P95, Conv1D recall on the majority anomaly type is only 0.22. The practical failure mode for Conv1D is not a rare class; it is that class-2 beats are too easy to reconstruct.

Table 2: Test recall by anomaly type at the calibration-F1 threshold.

Model	R-on-T PVC (353)	PVC (19)	SP/EB (39)	Uncl. (5)
LSTM AE	0.980	0.895	0.949	1.00
Conv1D AE	0.969	0.947	0.872	1.00

Failure cases. Figures 7 and 8 show the worst false positives and false negatives at the calibration-F1 cut (original vs. reconstruction). LSTM false positives are normal beats with unusual amplitude or timing that the 48-D bottleneck cannot copy; false negatives are abnormal beats whose reconstruction is still close. Conv1D false positives are numerous: many ordinary-looking normals sit above the low F1 threshold. That matches Table 1 — Conv1D’s F1 problem is a noisy score, not a handful of exotic morphologies.

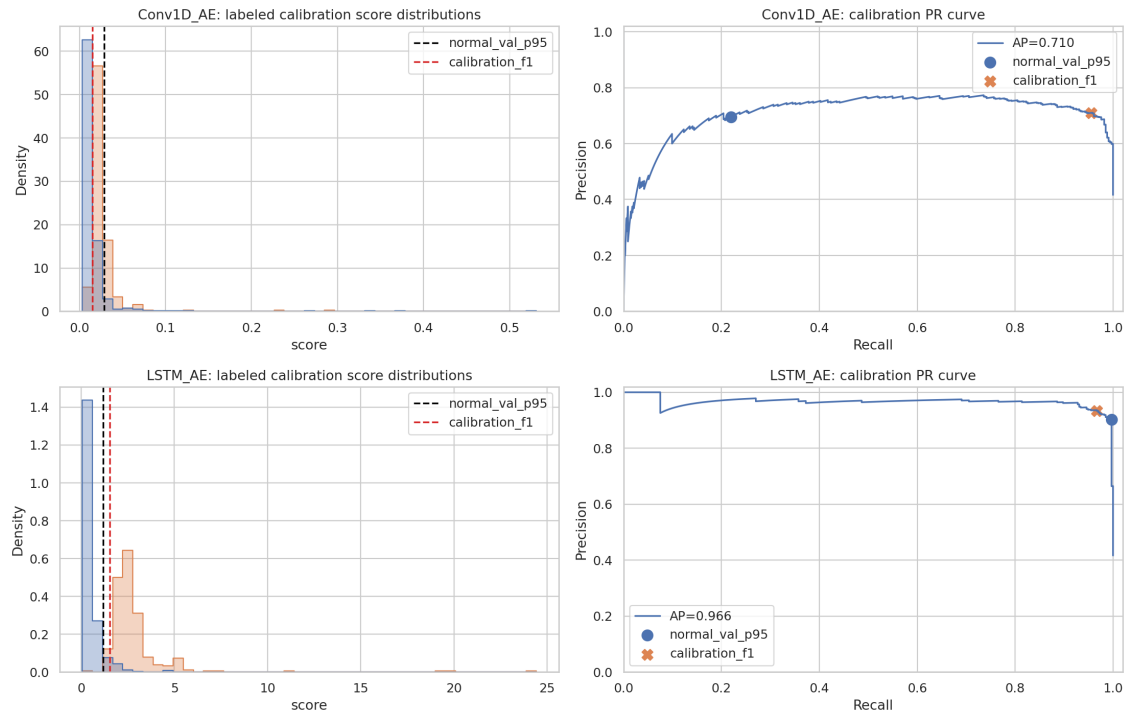


Figure 3: Score histograms on the labeled calibration split and the two frozen thresholds. Final-test labels are not used here.

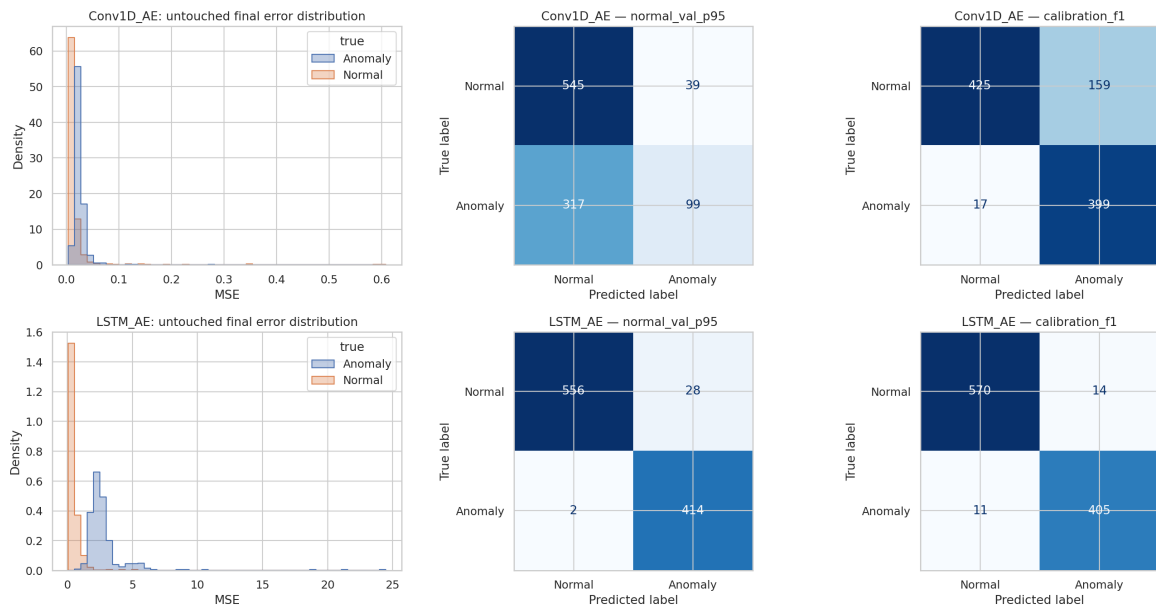


Figure 4: Test reconstruction-error distributions and confusion matrices at the calibration-F1 operating point.

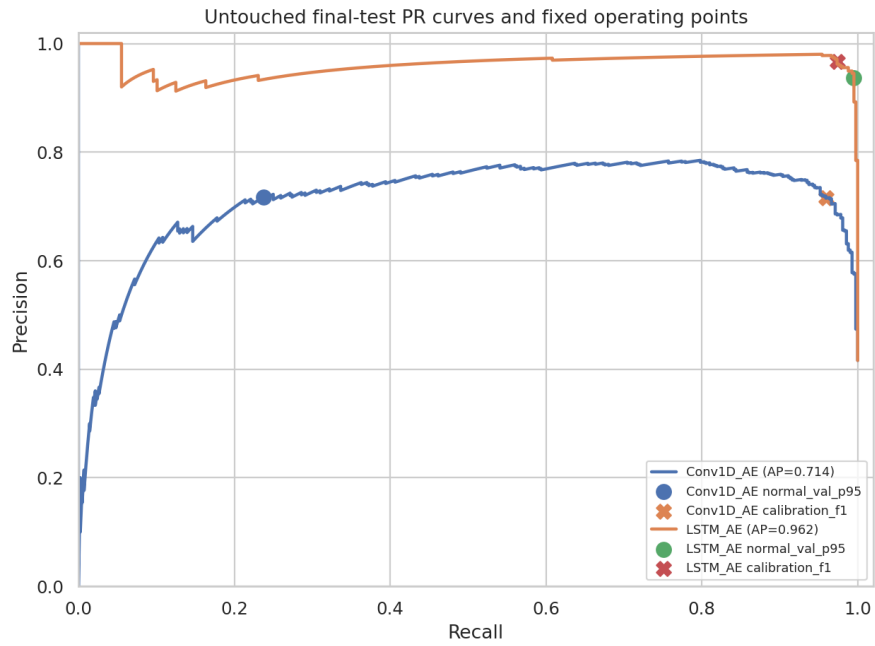


Figure 5: Test precision–recall curves. Marked points are the preselected thresholds, not oracle cuts on test.

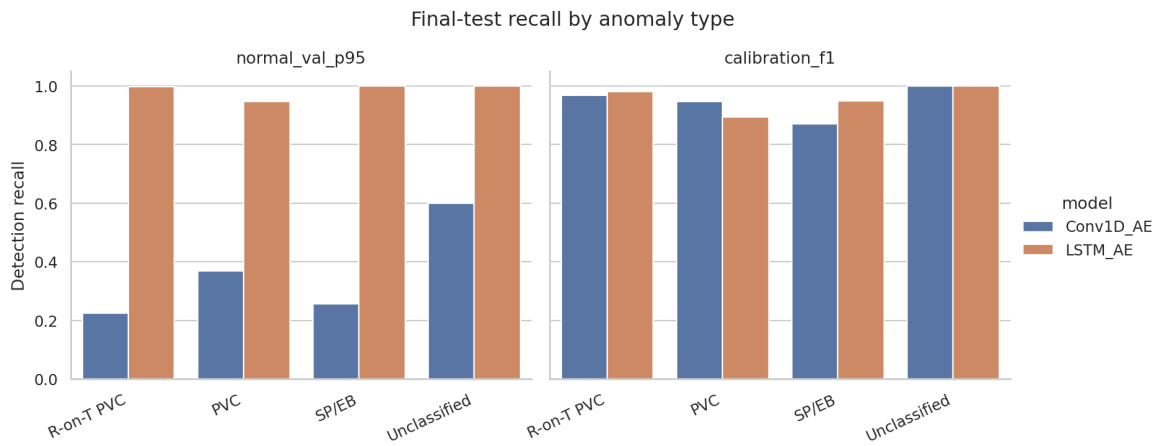


Figure 6: Per-type recall at both preselected thresholds. P95 is usable for LSTM and unusable for Conv1D.

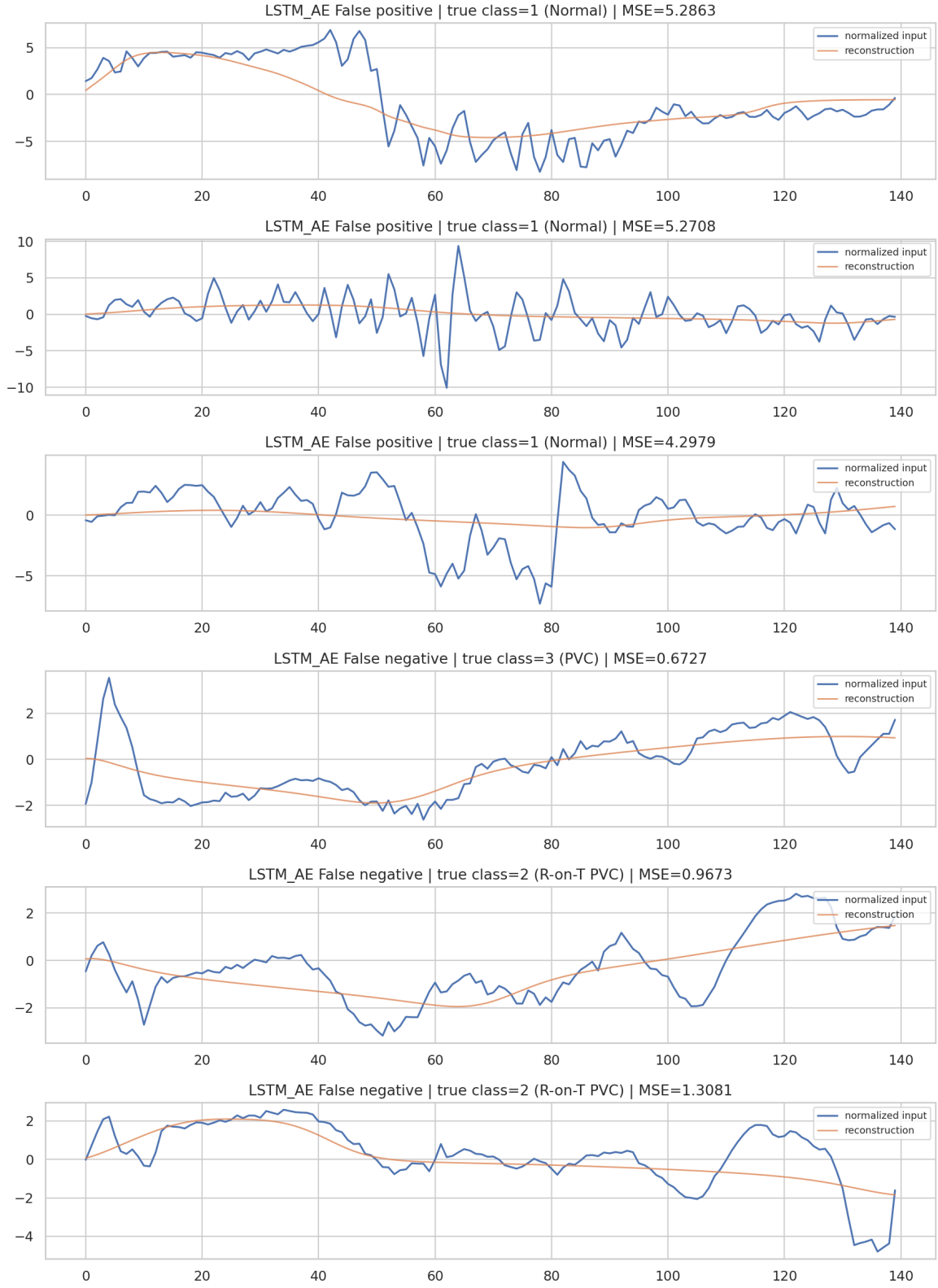


Figure 7: LSTM AE: highest-error false positives and false negatives (calibration-F1 threshold).

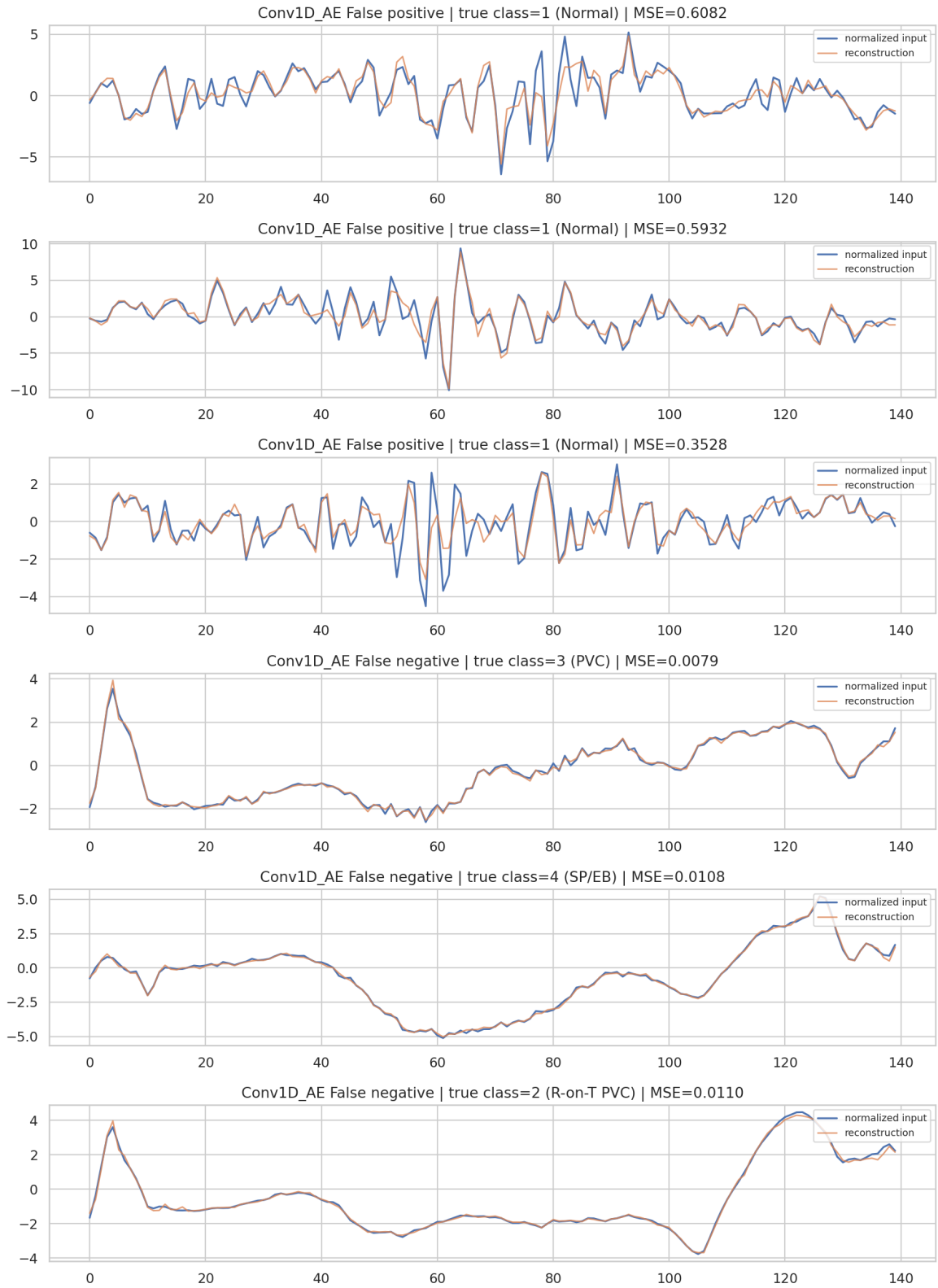


Figure 8: Conv1D AE: same selection rule. The false-positive panel is the operational issue.

Stretch baselines. One-Class SVM (RBF, $\nu = 0.05$) and Isolation Forest are fit on the same normalized normal-train matrix; their scores are oriented so larger means more anomalous, then cut with the same calibration-F1 recipe. SVM F1 0.972 slightly beats LSTM 0.970, with 3 FN vs. 11. LSTM still leads AP and AUC. Isolation Forest is clearly weaker (F1 0.940, AP 0.866). On this small, already-segmented benchmark a kernel one-class SVM is a strong baseline; the LSTM is competitive with it and far above Conv1D. We would not claim a neural win on F1.

5 Discussion and observation

The two questions below are from the project card [1].

Question 1

Reconstruction-based anomaly detection has a paradox: if the autoencoder becomes too good, it may reconstruct anomalies well and stop flagging them. Discuss this tension, and how you would deliberately limit the model (capacity, bottleneck size, training data) to keep it sensitive.

Answer. The detector assumes: normals lie on a low-dimensional manifold the AE can copy; anomalies do not. If the decoder is a near-universal interpolator, that assumption dies. An abnormal QRS can still be emitted sample-by-sample if the latent code and the decoder have enough degrees of freedom. The alarm score then looks “normal.”

This run is a concrete instance. The Conv1D latent map is 35×16 — 560 numbers to reconstruct 140 samples, i.e. not a tight bottleneck. Combined with shift-tolerant local filters, it can paste together a plausible beat even when the global shape is wrong. Hence AP only 0.71 and a P95 recall of 0.24: anomalies are reconstructed well enough to hide under the normal tail. The LSTM is limited to a 48-D summary of the whole beat. That is enough to copy the dominant normal template and not enough to freely redraw a PVC. Score separation follows (AP 0.96).

How we would keep a model sensitive, on purpose:

- **Bottleneck.** Prefer a vector whose dimension is well below 140 (here 48 worked; 8–16 would be the next experiment). Avoid overcomplete conv latents unless they are combined with a stronger penalty.
- **Capacity.** Fewer filters/units, no deep residual decoder. Stop when normal-validation loss plateaus; do not chase zero MSE.
- **Training data.** Train on normals only — already done. Adding even a few anomalies to the reconstruction loss would teach the model to copy them. A follow-up that *would* use anomalies is a contrastive or energy penalty, not MSE on the rare class.
- **Score.** MSE over the whole beat can be dominated by easy baseline wander. A QRS-windowed or residual-shape score would stay sensitive after a very good decoder appears.

The tension is not “never use a strong model.” It is: the object we optimize (copy normals) is not the object we need (fail to copy anomalies). Capacity must be enough for the first and insufficient for the second.

Question 2

If you added self-attention to the encoder, what might the attention weights reveal about which part of a heartbeat carries the “normal” signature — and would attention genuinely help, or just add capacity?

Answer. A normal ECG beat is not uniformly informative. The QRS complex (around the middle of these 140 samples) carries the spike timing and morphology that distinguish many of the labeled anomalies; P and T waves are lower amplitude and more variable. Self-attention over time could, in principle, put mass on QRS samples when reconstructing, and the heatmap would be a diagnostic: “this is what the encoder treats as the normal signature.” On a false negative, you could check whether attention still locked onto a QRS-shaped region while the true beat was ectopic — a useful failure story, similar to looking at LSTM reconstructions in Figure 7.

Would it help detection? Not automatically. Attention adds pairwise capacity. That can *worsen* the paradox in Q1: a flexible encoder is better at explaining away an abnormal beat as a reweighted combination of normal parts. On this dataset a 48-D LSTM already reaches AP 0.96 and a kernel SVM matches F1 without any sequence model. Extra attention is likely to be more capacity unless it is paired with a still-tight bottleneck and judged on held-out AP/F1, not on whether the heatmap looks anatomical. I would add a single attention pooling over the conv or LSTM states, freeze the same splits, and only keep it if test AP rises without flattening the anomaly-score tail. A pretty QRS heatmap alone would not count as a better detector.

6 Limitations

- Beats are short, univariate, and already segmented. Results will not transfer to raw multi-lead streams.
- Split is beat-level. Patient IDs are absent, so same-patient leakage is possible.
- One seed, one split. Architectures are protocol-matched, not parameter-matched (LSTM has $3.6\times$ Conv1D’s weights).
- F1 treats precision and recall equally and depends on the 42% anomaly rate in this test split; a screening clinic would not.
- Class names are archive morphology tags, not a clinical diagnosis. This is not a medical device.

7 Conclusion

On ECG5000, a 48-D LSTM autoencoder trained only on normal beats, with an F1 threshold fit on a separate labeled calibration set, detects anomalies at test F1 0.970 (precision 0.967, recall 0.974). A smaller Conv1D autoencoder is the wrong inductive bias here: it reconstructs too many abnormal beats, so an unsupervised P95 threshold fails (recall 0.24) and a labeled threshold buys recall with 159 false positives. One-Class SVM is a slightly stronger F1 baseline (0.972); LSTM remains the better neural model and the better ranked score. The useful original result is not “autoencoders work,” it is that bottleneck tightness and threshold source change the operating point more than the training loss does.

References

- [1] H. Farahani. *Data Mining Deep Learning Project Handbook*, Spring 2026. Project card AE-1.
- [2] H. A. Dau, A. Bagnall, K. Kamgar, C.-C. M. Yeh, Y. Zhu, S. Gharghabi, C. A. Ratanamahatana, and E. Keogh. The UCR time series archive. *IEEE/CAA J. Automatica Sinica*, 6(6):1293–1305, 2019. Dataset: ECG5000.
- [3] A. L. Goldberger, L. A. N. Amaral, L. Glass, J. M. Hausdorff, P. C. Ivanov, R. G. Mark, J. E. Mietus, G. B. Moody, C.-K. Peng, and H. E. Stanley. PhysioBank, PhysioToolkit, and PhysioNet: components of a new research resource for complex physiologic signals. *Circulation*, 101(23):e215–e220, 2000.